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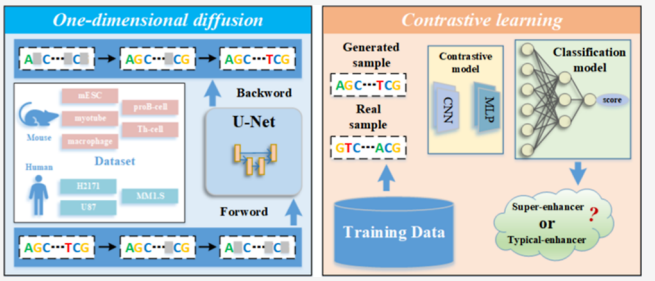
Diff-SE: A Diffusion-Augmented Contrastive Learning Framework for Super-Enhancer Prediction

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ABSTRACT: Super-enhancers (SEs) are cis-regulatory elements that play crucial roles in gene expression and are implicated in diseases such as cancer and Alzheimer's. Traditional identification methods rely on ChIP-seq experiments, which are costly and time-consuming. While recent computational approaches have leveraged sequence features for SE prediction, they often suffer from severe class imbalance and poor generalization across species. To address these limitations, we propose Diff-SE, a deep learning framework that integrates diffusion-based data augmentation with contrastive learning. The diffusion module models the continuous distribution of SEs to generate biologically meaningful synthetic positive samples, effectively balancing training data. A contrastive learning strategy is then used to enhance feature representation by maximizing intraclass similarity and interclass separation. Experimental results across eight data sets demonstrate that Diff-SE consistently outperforms the baseline model, achieving 10%−30% improvements in precision (PRE), Matthews correlation coefficient (MCC), and *F*1-score. Furthermore, Diff-SE exhibits superior generalization in cross-species validation between human and mouse cell lines. The code and data sets are available at https://github.com/15831959673/Diff-SE, enabling further research and applications in SE prediction.



1. INTRODUCTION

Super-enhancers (SEs) are cis-regulatory elements with exceptionally strong transcriptional activation properties.[1] Its regions typically span a range of 8000−20,000, while the span of typical enhancers (TEs) is between 200 and 300. Accurately predicting the ultralong SE sequences aids in research into mutations associated with various diseases such as cancer and Alzheimer's disease.[2] Wet lab experiments are one of the common methods to distinguish SEs. For instance, Wang et al. utilized BRD4 (a human bromodomain-containing protein) to discover SEs.[3] Jiang et al. created a catalog of SEs for 86 human cell and tissue samples depending on H3K27ac.[4] However, these approaches need to locate SEs based on the CHIP-seq,[5] which involve high costs on time.

With the development of artificial intelligence, machine learning or deep learning based on large-scale biological data has been employed to discover regulatory elements such as SEs.[6−9] Khan et al. developed a machine learning method called Improse, which combines 32 types of ChIP-seq features and 3 sequence features to uncover SEs and their components.[10] DeepSEN selected 36 epigenetic features and then applied convolutional neural networks to distinguish between SEs and TEs.[11] Those methods rely on ChIP-seq data from different factors to achieve high predictive performance. The shortcomings of the above methods are that they first require the features derived from gene sequencing to be established.[12]

Therefore, the following methods exclude experimental features other than DNA sequences, relying solely on sequence features to distinguish SEs from TEs, similar to the DeepSE approach.[13] Although this method demonstrates good performance in SE prediction tasks, it fails to effectively address the issue of extreme data imbalance. DeepSE supplements positive samples using the SMOTE algorithm;[14] however, this straightforward algorithm often struggles to generate samples with high-dimensional sequence feature information, resulting in insufficient focus on the prediction of positive samples within the DeepSE model. Additionally, the features employed by DeepSE delineate interspecies differences, which limits its generalization to species such as humans and mice.

In this study, we propose the Diff-SE model for the classification of DNA sequences into SEs and TEs. The Diff-SE model utilizes DNA2vec[7] and handcrafted features[15] to represent DNA sequences, and subsequently employs positive samples to train a diffusion model to generate high-quality

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synthetic positive samples, thereby balancing the data set. Finally, the balanced data set is used to train a downstream contrastive learning classification model. To validate the effectiveness of Diff-SE, we conducted baseline comparative experiments, ablation studies, and cross-species validation experiments, relying on publicly available data sets from humans and mice. The results demonstrate that our proposed method significantly enhances the performance of SE identification. Compared to the baseline DeepSE, Diff-SE achieved the best results across all metrics on seven data sets, with the exception of myotube cells, and most metrics showed an improvement in precision (PRE) of over 0.2. Furthermore, we validated the efficacy of Diff-SE in cross-species validation, with results that greatly surpassed those of DeepSE.

2. MATERIALS AND METHODS

2.1. Data Set. The data set utilized in this study encompasses eight data sets of SE sequences from both mice and humans. The mouse data sets include various cell types: embryonic stem cells (mESC), myotubes, macrophages, pro-B cells, and Th cells. The human data sets consist of three widely studied human cell types: small cell lung carcinoma (H2171), glioblastoma (U87), and multiple myeloma (MM1.S).[16] In these data sets, SEs are classified as positive samples, while TE is designated as negative samples. Each data set provides the genomic start and end coordinates for each enhancer, along with labels for positive and negative samples. Given the constraints of sequence feature representation and the significant differences in data length, we excluded sample shorter than 10bp. Detailed data information is presented in Table 1.

Table 1. Datasets for Different Species

species	cell line	TE	SE	SE/TE
mouse	mESC	8562	231	1:37
	myotube	4769	535	1:9
	macrophage	9514	961	1:10
	pro-B cell	13,419	396	1:34
	Th-cell	18,129	436	1:41
human	MM1.S	11,685	640	1:18
	H2171	16,354	357	1:46
	U87	19,231	1073	1:18

2.2. Methodology. Figure 1 illustrates the overall architecture of Diff-SE. As shown in Figure 1A, the limited number of positive samples and the imbalance in the data set contribute to a decline in performance metrics.[17] The proposed model comprises three main modules. Figure 1B illustrates the sequence feature representation module, which transforms SE sequences into feature vectors to support subsequent processing. The experimental section explains the embedding methodology in detail. Figure 1C presents the one-dimensional diffusion-based data augmentation module. This module fits a diffusion model using the embedded features and produces high-quality synthetic SE sequences. The model combines these synthetic samples with real SE, and down-samples the TE samples to form a balanced training data set, as described in Section 2.2.2. Figure 1D shows the contrastive learning classification module. This module applies contrastive learning to the balanced data set, trains a classification model, and produces predicted scores for samples.

2.2.1. Sequence Feature Representation. This section introduces four distinct types of feature embedding. The first is DNA2vec.[17] DNA2vec is a high-dimensional word embedding learned from DNA sequences based on the fundamental principles of word2vec. Compared to conventional vector embedding, DNA2vec captures deeper linguistic representations of DNA. However, it lacks the capacity to represent simple, surface-level sequence features. The second is k-mer frequency features,[18] K-mer embedding represent shallow-level sequence arrangement information, are easy to extract, and effectively capture fragment frequency patterns. However, they do not account for the correlation between sequential positions in the DNA. The third and fourth are SC-PseDNC,[19] and SC-PseTNC,[20] respectively. These two representations incorporate both dinucleotide and trinucleotide frequency information while simultaneously capturing sequence-order correlations. Furthermore, they also account for the physicochemical properties of DNA sequences, thereby complementing the limitations of the aforementioned embedding methods. Below, we will describe the construction methods of these feature representations in detail.

2.2.1.1. k-Mer Representations Based on DNA2Vec. Different from general k-mer representations, we divide DNA sequences into subsequences of length k and then use DNA2vec to embed these k-length subsequences (k-mer) into 100-dimensional vectors. This method obtains the semantic relations between subsequences and understands the local knowledge of DNA sequences, such as positional information and contextual associations. For a given DNA sequence of length s, it is defined as follows

$$s = n_1 n_2 \ldots n_l \tag{1}$$

where n_i (i = 1,2,...,l)represents the i-th nucleotide in the sequence. This sequence is split based on the value of k to obtain subsequences, which is denoted as $r(j,s)$with length of k to indicate the j-th subsequence. The set of these subsequences can be represented as

$$r(s) = [r(1, s), r(2, s) \ldots r(m, s)] \tag{2}$$

Through the operation of flooring, we separate the DNA sequence of length l into m ($m = l/k$) subsequences of length k. All subsequences $r(j,s)$ are converted into 100-dimension word embeddings by DNA2vec, defined as $f(r(j,s))$. At this point, the sequence s is turned into

$$e(s) = [f(r(1, s)), f(r(2, s)) \ldots f(r(m, s))] \tag{3}$$

Next, we set k to 4, 5, and 6, leading to $e(s)_{k=4}$, $e(s)_{k=5}$, and $e(s)_{k=6}$, respectively. Subsequently, we sum all subsequences within these sequences. Taking $e(s)_{k=4}$ as an example, this involves summing all subsequences $f(r(j,s))$, resulting in $\sum_{k=4} f(r(j, s))$ with vector dimensions of 1 × 100. These 100-dimension vectors from different k values are concatenated (marked with $\oplus$) to form the final representation $E(s)$ with a dimension of 1 × 300 for DNA sequence s.

$$E(s) = \sum_{k=4} f(r(i, s)) \oplus \sum_{k=5} f(r(i, s)) \oplus \sum_{k=6} f(r(i, s)) \tag{4}$$

The $E(s)$ in eq 4 refers to the first type of feature representation that we have extracted, which signifies the global information on the sequence from different k values. For example, k = 3 and k = 5 can respectively capture biological characteristics of amino acids and short peptides.

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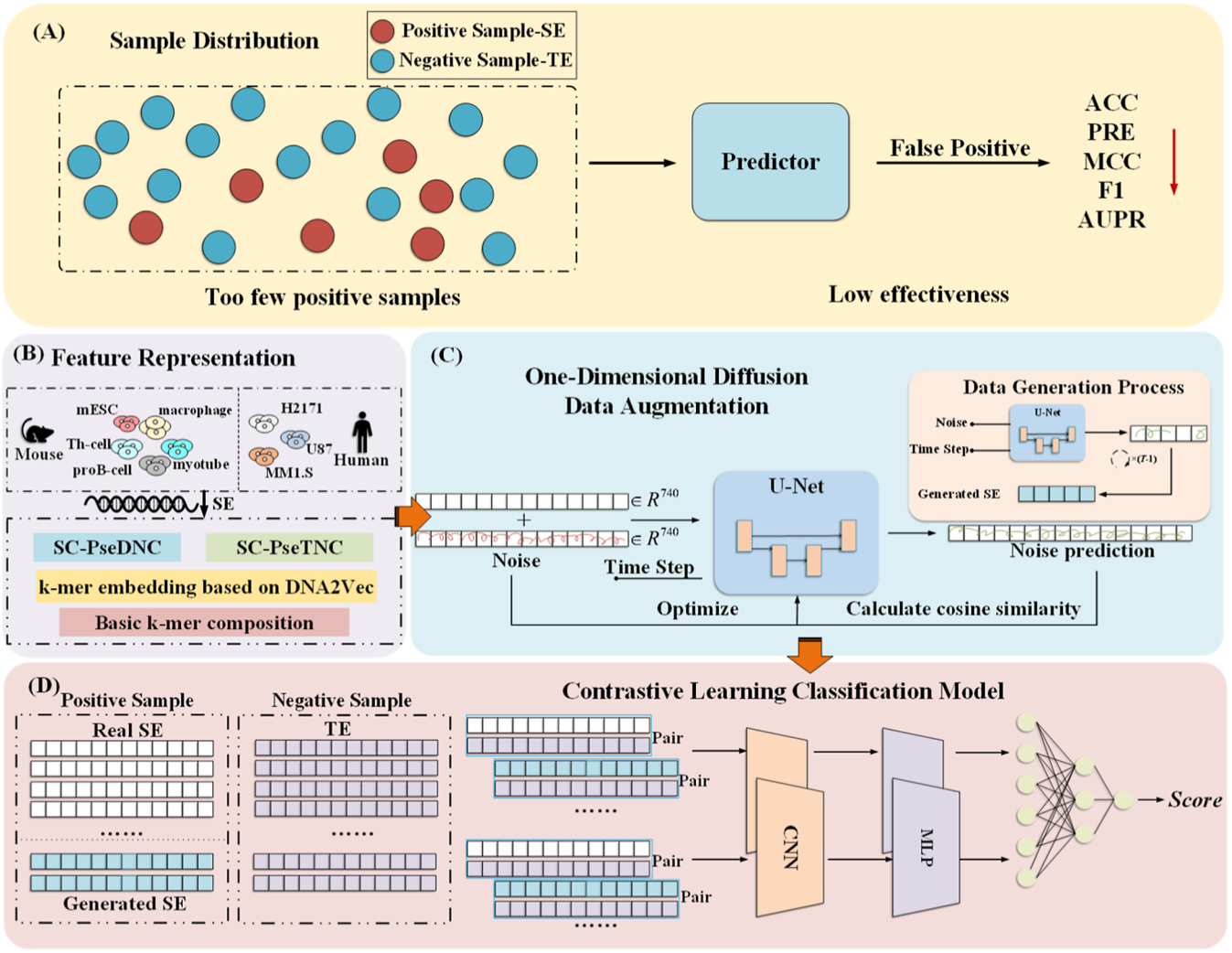


Figure 1. Framework of Diff-SE.

2.2.1.2. Basic k-mer Composition. Basic *k*-mer is the simplest approach to represent the DNAs, in which the DNA sequences are represented as the occurrence frequencies of k neighboring nucleic acids. This approach has been successfully applied to human gene regulatory sequence prediction,[21] enhancer identification.[22]

2.2.1.3. SC-PseDNC. The calculation method for SC-PseDNC (series correlation pseudo dinucleotide composition) is similar to that of SC-PseTNC.[23] Here, we introduce the principle using SC-PseDNC as an example. The SC-PseDNC feature vector of D is defined

$$D = [d_1, d_2 \ldots d_{16}, d_{16+1} \ldots d_{16+\lambda}, d_{16+\lambda+1} \ldots d_{16+\lambda\Lambda}]^T \quad (5)$$

$$d_k = \begin{cases} \dfrac{f_k}{\sum_{i=1}^{16} f_i + w\sum_{j=1}^{\lambda}\theta_j} \ (1 \le k \le 16) \\ \dfrac{w\theta_{k-16}}{\sum_{i=1}^{16} f_i + w\sum_{j=1}^{\lambda\Lambda}\theta_j} \ (17 \le k \le 16 + \lambda\Lambda) \end{cases} \quad (6)$$

where f_k $(k = 1,2...16)$ is the normalized occurrence frequency of dinucleotide in the DNA sequence; the parameter λ is an integer, representing the highest counted rank (or tier) of the correlation along a DNA sequence; w is the weight factor ranged from 0 to 1; Λ is the number of physicochemical indices; θ_j $(j = 1,2...\lambda)$ is called the j-tier correlation factor that reflects the sequence-order correlation between all the most contiguous dinucleotides along a DNA sequence, which is defined

$$\begin{cases} \theta_1 = \dfrac{1}{L-3}\sum_{i=1}^{L-3} J_{i,i+1}^1 \\ \theta_2 = \dfrac{1}{L-3}\sum_{i=1}^{L-3} J_{i,i+1}^2 \\ \ldots \\ \theta_\Lambda = \dfrac{1}{L-3}\sum_{i=1}^{L-3} J_{i,i+1}^\Lambda \lambda < (L-2) \\ \ldots \\ \theta_{\lambda\Lambda-1} = \dfrac{1}{L-\lambda-2}\sum_{i=1}^{L-\lambda-2} J_{i,i+\lambda}^{\Lambda-1} \\ \theta_{\lambda\Lambda} = \dfrac{1}{L-\lambda-2}\sum_{i=1}^{L-\lambda-2} J_{i,i+\lambda}^{\Lambda} \end{cases} \quad (7)$$

The correlation function is given by

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$$\begin{cases} J_{i,i+m}^{\zeta} = P_u(R_iR_{i+1}) \cdot P_u(R_{i+m}R_{i+m+1}) \\ \zeta = 1, 2 \ldots \Lambda;\ m = 1, 2 \ldots \lambda;\ i = 1, 2 \ldots L - \lambda - 2 \end{cases} \quad (8)$$

where μ is the number of total physiochemical indices; $P_u(R_iR_{i+1})$ represents the numerical value of the u-th ($u = 1,2\cdots\mu$) physiochemical index for the dinucleotide $R_iR_{i+1}(R_jR_{j+1})$ at position $i(j)$.

2.2.2. One-Dimensional Diffusion Data Augmentation. The augmentation of limited positive samples in DeepSE uses the SMOTE algorithm, which expands the minority class by generating synthetic samples within the feature space. The algorithm identifies the k nearest neighbors for each minority sample and selects a random point between the sample and one of its neighbors to produce a new instance.[24] This method demonstrates effectiveness in many scenarios. However, its performance depends on whether the available positive samples sufficiently represent the full pattern diversity of the positive class within the data set of the specific cell line. For instance, in the mESC cell line, there are only 231 positive samples out of a total of 8793 samples, resulting in an insufficient number of positive samples to capture the patterns of all super-enhancer (SE) categories within that cell line. Therefore, the diffusion model, which possesses the ability to generate diverse categories,[25] can significantly enhance the quality of the augmented data by better capturing the underlying diverse patterns present in the SE data set.

The diffusion model is a generative model that is widely applied in various generation fields.[26−28] The model comprises two main components: the forward noise addition process and the reverse denoising process. The forward process is a deterministic procedure that is intentionally computed, while the training of the diffusion model involves progressively training the model to learn the reverse denoising process based on the principle of stepwise noise addition. This approach facilitates data generation. The fundamental concept is to gradually add noise to the data until a standard normal distribution is formed, after which the model is trained to recover the original data from this distribution.[29]

In the context of this study, the diffusion model is required to generate SE samples. Let x_0 represents the feature vector of one real SE sample. Any step in the forward noise addition process of diffusion can be described as follows

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t;\ \sqrt{1-\beta_t}x_{t-1},\ \beta_t I) \quad (9)$$

In this equation, β_t is the hyperparameter that controls the amount of noise added at each step. Since this formula represents the forward noising process, x_{t-1} from the previous step is known, and x_t can be computed accordingly. After T steps, the data distribution approaches

$$q(x_T|x_0) = \mathcal{N}(x_T;\ 0,\ I) \quad (10)$$

The goal during the training phase is to learn the reverse process $p(x_{t-1}|x_t)$, which represents any step in the reverse denoising process. Its form is given by

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1};\ \mu_\theta(x_t, t),\ \sigma_\theta^2(x_t, t)I) \quad (11)$$

The terms containing θ need to be fitted through training a neural network, specifically training a U-net to fit the noise added in the forward process. The loss function is given by

$$L(\theta) = \mathbb{E}_{t,x_0,\epsilon}[\|\epsilon - \epsilon_\theta(x_t, t)\|^2] \quad (12)$$

In the generation phase, the process starts from a standard normal distribution $x_T \sim \mathcal{N}(0, I)$, and through the reverse process, noise is gradually removed, ultimately generating a new data sample x_0.

2.2.3. Contrastive Learning Classification. After generating data using the diffusion model, we obtained two data sets. The first part consists of the positive sample data set generated by the diffusion model, while the second part includes the original data set containing both positive and negative samples. Given the extreme imbalance between positive and negative samples, we generated 10,000 positive samples for each cell line. Subsequently, for each cell line, we constructed a training set by selecting a subset of generated samples equal to nine times the number of original positive samples. However, after combining these data sets, the total number of positive samples was still insufficient to balance the negative samples. Therefore, we performed random down-sampling on the negative samples to equalize their quantity with the positive samples, ultimately obtaining the negative sample set for the training data set. After this process, the positive and negative samples were combined to form the final training set for the classification model, as shown in Figure 1D.

Based on the constructed data set, we employed contrastive learning as the downstream classification model. Due to the extreme class imbalance in the data set, using traditional classification models would make it difficult to effectively learn the features of positive samples, leading to decreased classification accuracy. To address this issue, we chose the contrastive learning approach to enhance the model's focus on positive samples.[30] Before inputting the data into the classifier, a final processing step is required. Since contrastive learning aims to learn the differences between the two types of samples, the input must be presented in pairs. We randomly selected samples from each batch in the data set, paired them, and assigned a contrastive learning label to each pair: if the two samples belong to the same class, the label is set to 0; if they belong to different classes, the label is set to 1, thereby generating paired input data.

Our downstream model consists of two parallel subnetworks, which share parameters and are trained synchronously, as shown in Figure 1D. Each pair of input samples is processed separately by the subnetworks, which consist of convolutional layers and a multilayer perceptron (MLP). After training, the subnetworks gradually learn to pull similar features closer and push dissimilar features further apart, thereby improving the accuracy of the final fully connected classification layer. Therefore, we use the loss function Loss 1 to train the subnetworks and apply Loss 2 + Loss 3 to optimize the fully connected classification layer. Given the feature embeddings of two DNA sequence inputs, x_{sample_1} and x_{sample_2}, they are passed through the subnetworks simultaneously, as follows

$$v_1 = \text{MLP}(\text{Relu}(\max\text{pool}(\text{Conv}(\max\text{pool}(\text{Conv}(x_{\text{sample}_1}))))))) \quad (13)$$

$$v_2 = \text{MLP}(\text{Relu}(\max\text{pool}(\text{Conv}(\max\text{pool}(\text{Conv}(x_{\text{sample}_2}))))))) \quad (14)$$

Here, v_1 and v_2 represent the high-dimensional feature representations obtained after passing through the subnetworks. The Euclidean distance loss is computed between them,

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Table 2. Comparison with Baseline Results under 10-Fold Cross Validation

species	cell line	model	ACC	PRE	MCC	*F*1
mouse	mESC	DeepSE	0.913	0.200	0.363	0.310
		Diff-SE	**0.969**	**0.458**	**0.447**	**0.448**
	myotube	DeepSE	0.898	0.50	0.50	0.551
		Diff-SE	**0.906**	**0.541**	0.488	0.537
	macrophage	DeepSE	0.862	0.386	0.501	0.524
		Diff-SE	**0.934**	**0.706**	**0.566**	**0.587**
	pro-B cell	DeepSE	0.877	0.168	0.338	0.279
		Diff-SE	**0.966**	**0.462**	**0.459**	**0.468**
	Th-cell	DeepSE	0.936	0.278	0.489	0.423
		Diff-SE	**0.985**	**0.696**	**0.688**	**0.690**
human	H2171	DeepSE	0.917	0.202	0.400	0.325
		Diff-SE	**0.982**	**0.604**	**0.579**	**0.582**
	U87	DeepSE	0.902	0.348	0.530	0.503
		Diff-SE	**0.965**	**0.667**	**0.655**	**0.672**
	MM1.S	DeepSE	0.897	0.328	0.501	0.477
		Diff-SE	**0.959**	**0.608**	**0.616**	**0.627**

aiming to minimize the distance for samples of the same class and maximize the distance for samples of different classes. The loss function Loss 1 is formulated as follows

$$\text{Loss } 1(v_1, v_2, Y) = \frac{1}{2}(1 - Y)d(v_1, v_2)^2 + \frac{1}{2}Y\{\max(0, m - d(v_1, v_2)^2)\} \tag{15}$$

Here, Y represents the contrastive learning label, and $d(v_1,v_2)$ denotes the Euclidean distance between v_1 and v_2. m is the threshold that triggers optimization; this term is only computed when $Y = 1$, indicating that the two samples do not belong to the same class. If the distance between v_1 and v_2 exceeds m, no optimization is performed. However, if the distance between v_1 and v_2 is less than m, it indicates that the distance between different classes is too small, necessitating an update of the model parameters to adjust the distance to as close as possible to m.

For the subsequent fully connected layer sequence classification task, we employ the binary cross-entropy (BCE) loss, denoted as Loss 2, which is formulated as follows

$$\text{Loss } 2\ (v_i, Y_i) = \sum_{i=1}^{2} [Y_i \log(p_{v_i}) + (1 - Y_i)\log(1 - p_{v_i})] \tag{16}$$

Here, Y_i represents the class label of the input sample, and p_{vi} denotes the classification probability of the input sample.

2.2.4. Implementations. The Diff-SE model is implemented using PyTorch (version 1.13.1) and trained on an NVIDIA 4090 GPU with a batch size of 32. The Adam optimizer is used, and hyperparameter tuning is performed using 10-fold cross-validation. The diffusion model training component of Diff-SE consumes approximately 15 GB of GPU memory and requires around 10 GPU hours for training. The downstream contrastive learning component uses about 1 GB of GPU memory and takes approximately 2 GPU hours to train. All hyperparameters of Diff-SE when trained on the benchmark data sets are presented in Tables S1−S6.

2.2.5. Metrics. We employed five metrics to evaluate the performance of Diff-SE: accuracy (ACC), precision (PRE), *F*1 score (*F*1), Matthews correlation coefficient (MCC), and area under the precision−recall curve (AUPR). *F*1, AUPR, and MCC serve as comprehensive evaluation metrics.[31,32] The *F*1 score represents the harmonic mean of precision and recall, and it is primarily used to assess the accuracy of binary classification models. This metric is particularly important in imbalanced data sets, as it takes into account the performance of both positive and negative classes.[33] AUPR quantifies the area under the precision−recall curve, which provides a better reflection of the model's performance when dealing with imbalanced data sets. MCC is a holistic metric that effectively captures the overall performance of the model in binary classification tasks by considering all four classification outcomes: true positives, false positives, true negatives, and false negatives. The value of MCC ranges from −1 to 1, where 1 indicates perfect classification, 0 represents random classification, and −1 signifies complete misclassification.

$$\text{ACC} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{17}$$

$$\text{PRE} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{18}$$

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP}) \times (\text{TP} + \text{FN}) \times (\text{TN} + \text{FP}) \times (\text{TN} + \text{FN})}} \tag{19}$$

$$F1 = \frac{2\text{PRE} \times \text{REC}}{\text{PRE} + \text{REC}} \tag{20}$$

3. EXPERIMENTS AND DISCUSSIONS

3.1. Performance of Diff-SE. To validate the performance of the Diff-SE model in SE detection, we selected the DeepSE model as a comparative baseline. We conducted 10-fold cross-validation on the eight data sets, using the average performance across the ten folds as the evaluation criterion. Given the imbalanced nature of the data sets, we applied the SMOTE method to generate positive samples for DeepSE as described in the original publication, while Diff-SE utilized a diffusion model to generate positive samples.

Table 2 presents the results of SE prediction across the eight data sets, highlighting the best metric values for each data set in bold. Except for the myotube data set, Diff-SE achieved superior results across all other seven data sets. For instance, in

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the H2171 cell line, Diff-SE outperformed DeepSE with an increase of 0.065 in accuracy (ACC), an improvement of 0.402 in precision (PRE), a rise of 0.179 in Matthews correlation coefficient (MCC), and an increase of 0.257 in *F*1 score. The performance of Diff-SE demonstrated significant enhancements across the other data sets as well, particularly in PRE, indicating that the model predicts a limited number of critical positive samples more accurately in imbalanced data sets. The improvements in *F*1 score and ACC suggest that while enhancing the prediction capability for positive samples, the model also effectively maintains its predictive ability for negative samples. Furthermore, the increase in MCC further substantiates the significant advantages of Diff-SE in handling classification tasks with imbalanced data. In the myotube data set, we did not achieve an overall improvement; however, the precision (PRE) value of Diff-SE increased by 0.41 compared to DeepSE, while the Matthews correlation coefficient (MCC) and *F*1 score only declined by approximately 0.1. This indicates that, although the increased focus on positive samples led to a slight reduction in MCC and *F*1 score, the significant enhancement in PRE is a worthwhile trade-off.

The analysis of the classification results across the eight data sets reveals that Diff-SE has achieved comprehensive and significant improvements compared to DeepSE. Additionally, Figure 2 illustrates the comparison of Diff-SE and DeepSE in terms of the area under the precision−recall curve (AUPR). The blue curve represents the AUPR values of Diff-SE across the eight data sets, while the red curve corresponds to DeepSE. It is evident from the figure that all red curves are located within the blue curves, indicating that Diff-SE outperforms DeepSE in terms of the AUPR metric as well.

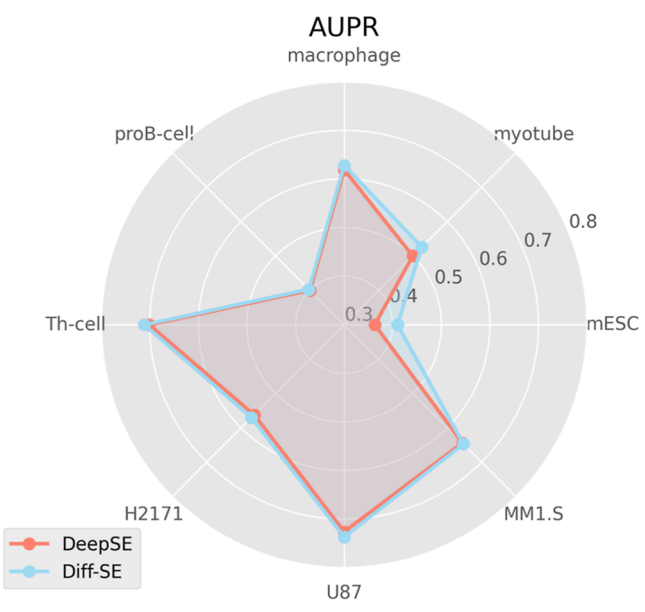


Figure 2. AUPR display of Diff-SE and DeepSE on eight data sets.

3.2. Ablation Study. In this section, we first trained a baseline CNN model[34,35] that outperformed DeepSE as the foundation model, and we designed ablation experiments to verify the roles of the diffusion model and contrastive learning in SE prediction. The specific experimental settings are as follows:

1 CNN-SMOTE: this model addresses the issue of data imbalance using the SMOTE method for SE prediction with a CNN model.
2 CNN-Diff: in this model, we use data generated by the diffusion model to create a balanced data set, which is then used for SE prediction with a CNN model.
3 Diff-SE: similarly, this model utilizes the balanced data set generated by the diffusion model, further integrating contrastive learning and fully connected layers for SE prediction.

Figure 3 illustrates the specific performances of these three models across the eight data sets. In all data sets except for myotube, the precision (PRE) metric for CNN-SMOTE is significantly lower than that of the other two models. When the diffusion model is employed instead of SMOTE to address the scarcity of positive samples, the model performance shows a notable improvement. This indicates that the quality of data generated by SMOTE is far inferior to that produced by the diffusion model, further validating the effectiveness of the diffusion model in tackling data imbalance tasks.

Moreover, by replacing CNN with contrastive learning in our proposed final model, we observe substantial improvements across all eight data sets compared to the first two models, particularly in the enhancement of the PRE metric. The increase in the PRE metric reflects a higher accuracy of the model in identifying positive samples; specifically, it indicates that among all samples predicted as positive, the proportion that is actually positive is significantly higher. This suggests that contrastive learning focuses more on learning the features of positive samples, achieving the intended effect we aimed for. The combination of the diffusion model and contrastive learning effectively enhances the model's attention to positive samples, leading to comprehensive improvements over conventional models.

3.3. Comparative Learning Analysis. Deep learning differs from other machine learning approaches that directly utilize handcrafted features, as the effectiveness of the latter's features is often more interpretable.[36] In contrast, the input features in deep learning undergo significant transformations as they are processed through various network layers, resulting in embeddings that exist in high-dimensional space. Consequently, feature visualization is frequently employed in deep learning to intuitively perceive the differences between features.

In Section 2.2.3, we analyzed the rationale for employing contrastive learning, which aims to achieve an effective separation of features between positive and negative samples. This process seeks to increase the disparity in the embedded positions of positive and negative sample features within the feature space. To this end, we utilized the t-SNE algorithm to reduce the dimensionality of the output from the contrastive learning subnet to two dimensions.[37] The purpose of this subnet is to amplify the differences between the two classes of features. Taking the mESC cell line as an example, Figure 4 illustrates the feature visualization after training within a 10-fold cross-validation framework. In the scatter plot, red points represent negative samples, while blue points represent positive samples. It is evident that these two classes of samples are indeed embedded in distinct positions within the two-dimensional space. This observation indicates that our contrastive learning network has successfully separated the features of the two categories, thereby laying a solid foundation

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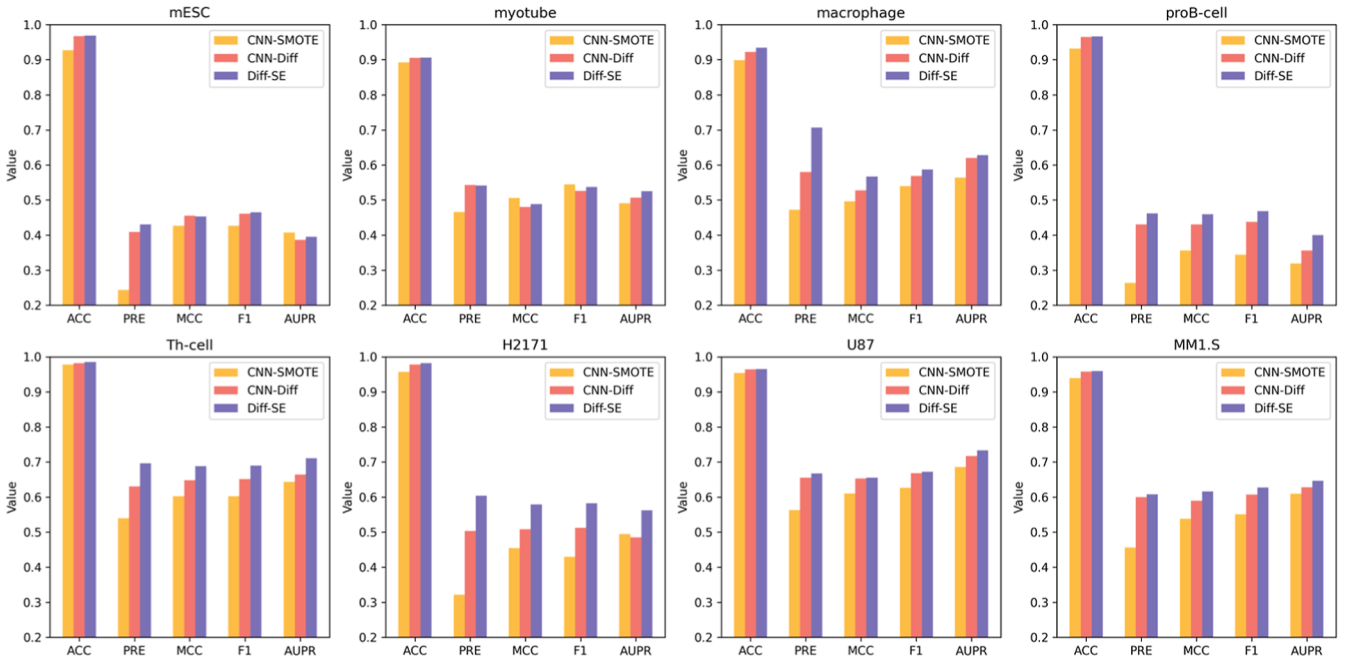


Figure 3. Comparison of prediction results of three ablation models on eight data sets.

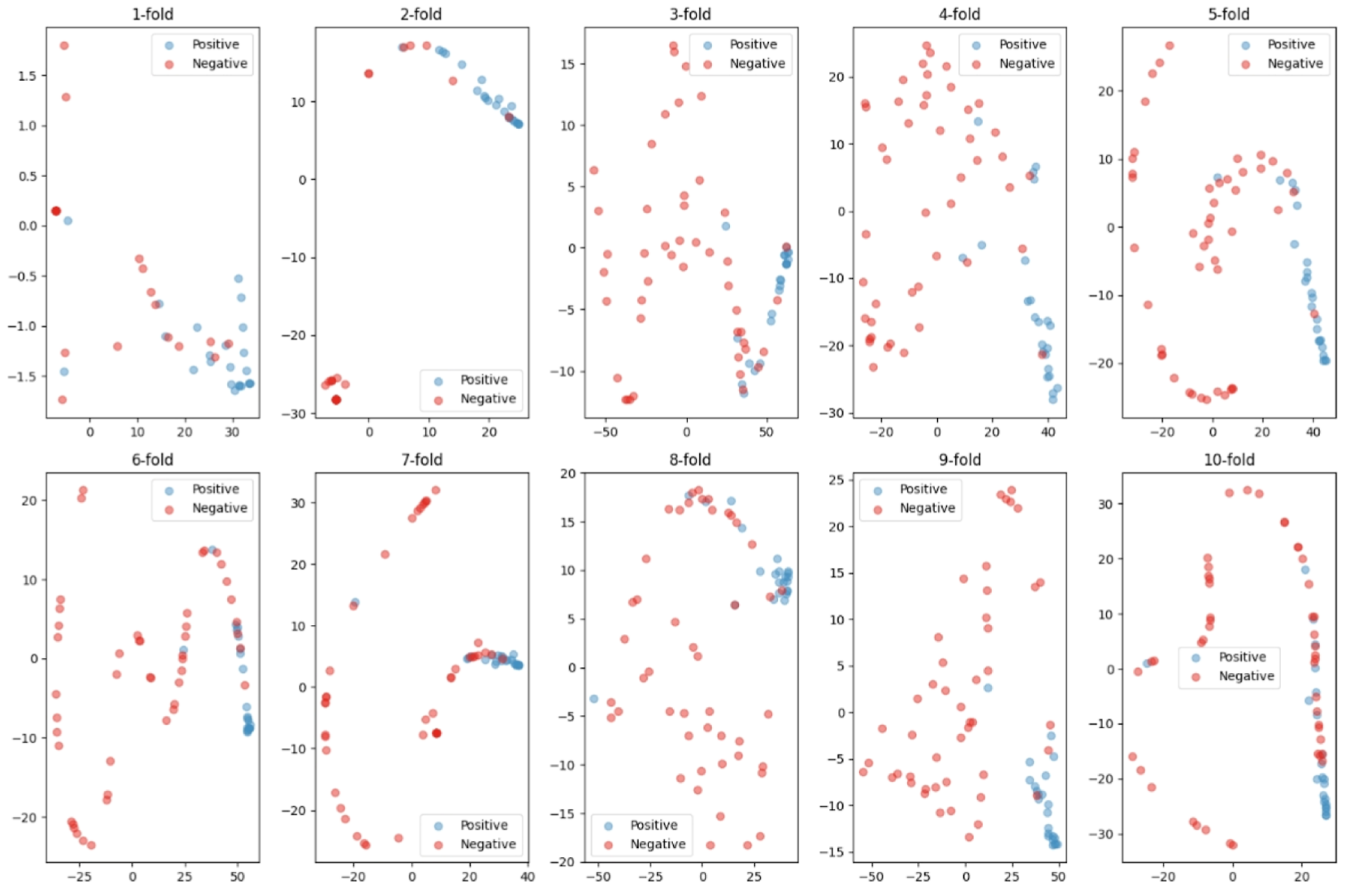


Figure 4. Visualization results of 10-fold comparison learning on mESC data set.

for the superior classification performance of downstream classifiers.

To further elucidate how contrastive learning effectively separates the high-dimensional features of SE and TE, we visualized the loss reduction throughout the training process. The contrastive loss incorporates a tunable hyperparameter, margin, which controls the threshold for initiating contrastive learning to update the model parameters. In this study, we set the margin to 2. When the vector distance between different class samples exceeds the margin, no processing occurs;

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conversely, when the vector distance is less than the margin, model parameters are updated to increase the distance between the two classes of samples after passing through the contrastive learning subnet.

In Figure 5, Loss 1 represents the contrastive learning loss, while the sum of Loss 2 and Loss 3 (the same as Loss 2) reflects the cross-entropy loss for downstream classification. Initially, it can be observed that the contrastive learning loss declines rapidly over the course of 10 epochs, subsequently tapering off, with stabilization occurring around 70 to 80 epochs. This trend indicates that contrastive learning effectively reduces the distance among samples of the same class while increasing the distance between samples of different classes. Furthermore, a comparison of the two loss graphs clearly shows that the predictive loss declines in tandem with the reduction in contrastive loss, providing further evidence that the feature embeddings obtained through contrastive learning significantly enhance the performance of the classification task.

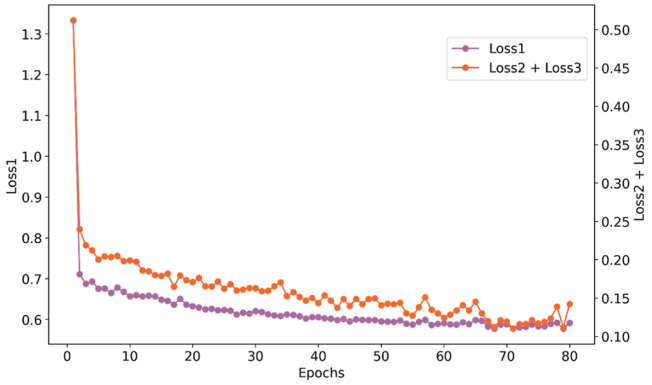


Figure 5. Loss curves for contrastive learning loss (Loss 1) and classification loss (Loss 2 + Loss 3).

3.4. Cross-Cell Line Validation. To improve generalizability beyond the same-cell-line setting in Section 3.1, we constructed a species-specific integrative model using data from all cell lines. We aggregated five mouse and three human cell line data sets and trained the models using stratified 10-fold cross-validation. We then evaluated performance on each individual cell line and compared it with the single-cell-line baseline from Section 3.1. As indicated in Table 3, the predictive performance of the integrative model was generally slightly lower than that of the baseline model in most cases. Compared to models trained on a single cell line, the integrated model exhibited slightly lower overall performance under cross-cell-line validation. This is likely due to the heterogeneity across cell lines, which introduces distribution shifts and reduces the effectiveness of cell-line-specific features when generalized across multiple contexts. However, compared to the DeepSE prediction results in Section 3.1, the performance of the integrative model still showed significant improvement. This suggests that the integrative model does not always outperform the baseline on individual cell lines. However, it effectively captures shared underlying features across different cell types. This makes it suitable for accurate predictions on each specific cell line.

Although the integrative model captures features from each cell line, it still does not validate whether a model trained on one cell line can generalize to others. Therefore, we conducted cross-validation at the individual cell line level. Figure 6 presents eight subplots. Each subplot is labeled by the training cell line, while the circular bar charts on the *X*-axis represent the testing cell lines. The mouse data sets include mESC, myotube, macrophage, proB-cell, and Th-cell, while the human data sets comprise H2171, U87, and MM1.S. It can be observed that when mouse species cell lines are used as the training set, the ACC values for all mouse cell lines exceed 80%. Despite a decrease in performance on human cell lines, the ACC values still remain above 75%. Conversely, when human cell lines serve as the training set, the ACC values exceed 90%, although the ACC values for mouse cell lines decline, they also remain above 75%. This indicates that our model is capable of capturing shared latent information across different cell lines. Within the same species, even though cell lines may originate from distinct tissues or developmental contexts, their core gene regulatory networks and epigenetic mechanisms tend to exhibit a high degree of conservation. In particular, regulatory elements associated with SE,[38] such as transcription factor (TF) binding sites and histone modifications (e.g., H3K27ac),[39] often display similar distribution patterns across multiple cell lines. The term "latent

Table 3. Performance Comparison between Integrative (Multi-Line) and Single-Cell-Line Training Models

species	cell line	model	ACC	PRE	MCC	*F*1
mouse	mESC	integrative	0.969	0.427	0.389	0.400
		single-cell-line	0.969	0.458	0.447	0.448
	myotube	integrative	0.911	0.611	0.433	0.459
		single-cell-line	0.906	0.541	0.488	0.537
	macrophage	integrative	0.930	0.657	0.533	0.563
		single-cell-line	0.934	0.706	0.566	0.587
	pro-B cell	integrative	0.966	0.440	0.442	0.452
		single-cell-line	0.966	0.462	0.459	0.468
	Th-cell	integrative	0.985	0.697	0.689	0.693
		single-cell-line	0.985	0.696	0.688	0.690
human	H2171	integrative	0.981	0.565	0.567	0.569
		single-cell-line	0.982	0.604	0.579	0.582
	U87	integrative	0.955	0.567	0.620	0.633
		single-cell-line	0.965	0.667	0.655	0.672
	MM1.S	integrative	0.956	0.589	0.559	0.577
		single-cell-line	0.959	0.608	0.616	0.627

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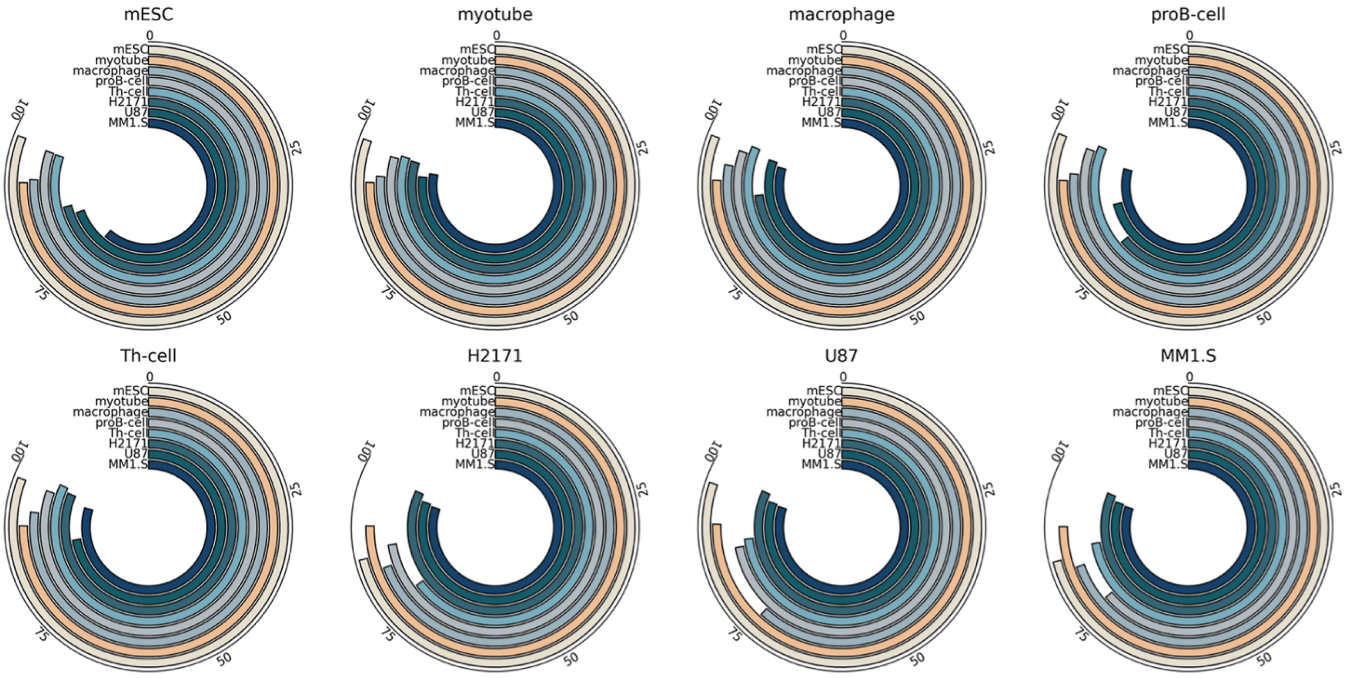


Figure 6. Accuracy of cross-cell-line validation using models trained on individual cell lines.

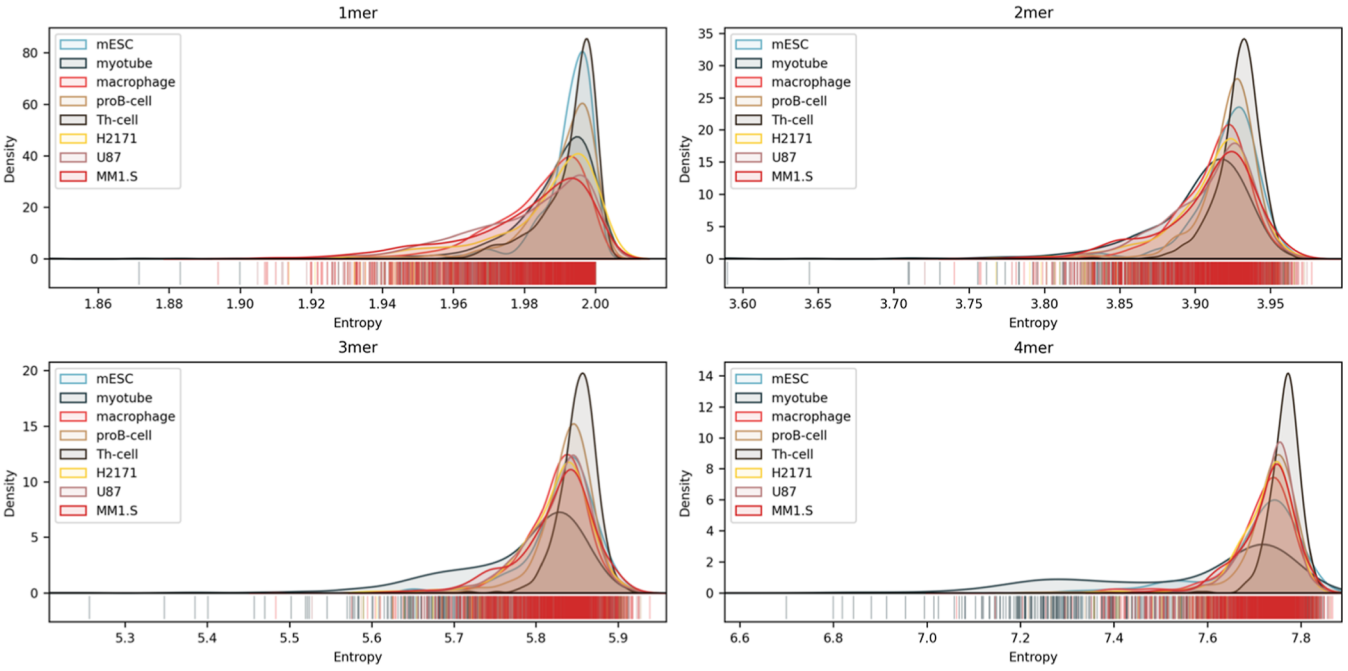


Figure 7. Kernel density plots of *k*-mer entropy values for different *k* values.

information" as referred to in this context encapsulates these underlying distribution patterns that are not readily discernible by human intuition from the raw DNA sequence but can be effectively learned and recognized by deep learning models.[40] It is precisely this similarity in distribution that enables the model to generalize well across cell lines of the same type. Moreover, there exists a greater abundance of latent information shared among cell lines within the same species, whereas such information becomes relatively sparse when comparing across species.

In summary, the results of these three experiments lead us to conclude that our model effectively captures latent features both between different species and among different cell lines within the same species. However, prior work such as DeepSE failed to achieve cross-validation at the species level. We argue that the cross-species information is primarily reflected in the manually extracted features. To support this, we visualized the frequency entropy of *k*-mers and plotted their kernel density estimates. As illustrated in Figure 7, it can be observed that the entropy distributions of 1-mer to 4-mer are concentrated at specific peaks. This suggests that there is a similar distribution of manually extracted *k*-mer features across different cell lines of different species, which provides strong support for the cross-validation experiments.

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3.5. Cross-Species Validation. To investigate whether there exists a cross-species shared information pattern among SE sequences across different species, we conducted a cross-species validation. The specific method involved integrating data sets from the same species to construct separate data sets for human and mouse species. Subsequently, we employed a stratified 10-fold sampling approach, alternating between mouse and human cell line data sets as training and testing sets for evaluation. In contrast to the description in DeepSE, our model successfully captured the shared information patterns between the two species. As shown in Table 4, the predictions from these two evaluations demonstrated good performance across different species.

Table 4. Species Cross-Validation Results between the Human and Mouse Datasets

train	test	ACC	PRE	MCC	*F*1
mouse	human	0.958	0.528	0.542	0.554
human	mouse	0.835	0.315	0.351	0.366

4. CONCLUSION

This study developed a model named Diff-SE, aimed at identifying whether a DNA sequence is a SE. To address the challenges posed by imbalanced data sets, we adapted a diffusion model to generate data of varying lengths, which has had a significant positive impact on downstream tasks. To enhance the model's focus on the characteristics of positive samples within imbalanced data sets, we employed a contrastive learning training strategy and investigated its effectiveness. Our model significantly improved upon the baseline in terms of generalization capabilities across different species. Within the same cell line, Diff-SE exhibited an enhancement of 10% to 30% in PRE, MCC, and *F*1 score across eight data sets compared to the baseline model. Furthermore, we conducted feature analysis to evaluate the model's generalization capabilities at the species level, revealing that k-mer features provided effective support in both species-level and cell line-level cross-validation. Diff-SE excels in super-enhancer prediction but has limitations: its 1D diffusion model is computationally expensive; contrastive learning requires sensitive hyperparameter tuning; and cross-species validation is limited to human-mouse, needing future tests on more distant species.

■ ASSOCIATED CONTENT

Data Availability Statement

The data and code of this work is accessible at https://github.com/15831959673/Diff-SE.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jcim.5c01005.

All hyperparameters of Diff-SE when trained on the benchmark data sets (PDF)

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Author Contributions

H.W.Y. and Z.H.L contributed equally to this work. Z.H.L and H.Y. performed the computational experiments. H.W.Y., Z.H.L, and B.Y.D. wrote and revised the manuscript. Z.Y. and G.F. conceived the study and supervised the research. All authors have given approval for the final version of the manuscript.

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Notes

The authors declare no competing financial interest.

■ REFERENCES

(1) Koutsi, M. A.; Pouliou, M.; Champezou, L.; Vatsellas, G.; Giannopoulou, A.-I.; Piperi, C.; Agelopoulos, M. Typical enhancers, super-enhancers, and cancers. *Cancers* **2022**, *14*, 4375.

(2) Kanzi, A. M.; San, J. E.; Chimukangara, B.; Wilkinson, E.; Fish, M.; Ramsuran, V.; De Oliveira, T. Next generation sequencing and bioinformatics analysis of family genetic inheritance. *Front. Genet.* **2020**, *11*, 544162.

(3) Wang, X.; Kutschat, A. P.; Yamada, M.; Prokakis, E.; Böttcher, P.; Tanaka, K.; Doki, Y.; Hamdan, F. H.; Johnsen, S. A. Bromodomain protein BRDT directs ΔNp63 function and super-enhancer activity in a subset of esophageal squamous cell carcinomas. *Cell Death Differ.* **2021**, *28*, 2207−2220.

(4) Beacon, T. H.; Delcuve, G. P.; López, C.; Nardocci, G.; Kovalchuk, I.; van Wijnen, A. J.; Davie, J. R. The dynamic broad epigenetic (H3K4me3, H3K27ac) domain as a mark of essential genes. *Clin. Epigenet.* **2021**, *13*, 138.

(5) Jia, Q.; Chen, S.; Tan, Y.; Li, Y.; Tang, F. Oncogenic super-enhancer formation in tumorigenesis and its molecular mechanisms. *Exp. Mol. Med.* **2020**, *52*, 713−723.

(6) He, W.; Xu, J.; Zuo, Y.; Bai, Y.; Guo, F. EnsembleSE: identification of super-enhancers based on ensemble learning. *Briefings Funct. Genomics* **2025**, *24*, No. elaf003.

(7) He, W.; Zhou, H.; Zuo, Y.; Bai, Y.; Guo, F. MuSE: A deep learning model based on multi-feature fusion for super-enhancer prediction. *Comput. Biol. Chem.* **2024**, *113*, 108282.

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(8) Guo, F.; Guan, R.; Li, Y.; Liu, Q.; Wang, X.; Yang, C.; Wang, J. Foundation models in bioinformatics. *Natl. Sci. Rev.* **2025**, *12*, nwaf028.
(9) Wei, Z.; Hua, K.; Wei, L.; Ma, S.; Jiang, R.; Zhang, X.; Li, Y.; Wong, W. H.; Wang, X. NeuronMotif: Deciphering cis-regulatory codes by layer-wise demixing of deep neural networks. *Proc. Natl. Acad. Sci. U.S.A.* **2023**, *120*, No. e2216698120.
(10) Khan, A.; Zhang, X. Integrative modeling reveals key chromatin and sequence signatures predicting super-enhancers. *Scientific Reports* **2019**, *9*, 2877.
(11) Bu, H.; Hao, J.; Gan, Y.; Zhou, S.; Guan, J. DEEPSEN: a convolutional neural network based method for super-enhancer prediction. *Bioinformatics* **2019**, *20*, 598.
(12) Su, Q.; Pham, N. T.; Wei, L.; Manavalan, B. MST-m6A: a novel multi-scale transformer-based framework for accurate prediction of m6A modification sites across diverse cellular contexts. *J. Mol. Biol.* **2025**, *437*, 168856.
(13) Ji, Q.-Y.; Gong, X.-J.; Li, H.-M.; Du, P.-F. DeepSE: Detecting super-enhancers among typical enhancers using only sequence feature embeddings. *Genomics* **2021**, *113*, 4052−4060.
(14) Kosolwattana, T.; Liu, C.; Hu, R.; Han, S.; Chen, H.; Lin, Y. A self-inspected adaptive SMOTE algorithm (SASMOTE) for highly imbalanced data classification in healthcare. *BioData Min.* **2023**, *16*, 15.
(15) Das, S.; Biswas, S. K.; Purkayastha, B. Automated Indian sign language recognition system by fusing deep and handcrafted feature. *Multimed. Tool. Appl.* **2023**, *82*, 16905−16927.
(16) Hamamoto, R.; Takasawa, K.; Shinkai, N.; Machino, H.; Kouno, N.; Asada, K.; Komatsu, M.; Kaneko, S. Analysis of super-enhancer using machine learning and its application to medical biology. *Briefings Bioinf.* **2023**, *24*, bbad107.
(17) Thölke, P.; Mantilla-Ramos, Y.-J.; Abdelhedi, H.; Maschke, C.; Dehgan, A.; Harel, Y.; Kemtur, A.; Mekki Berrada, L.; Sahraoui, M.; Young, T.; et al. Class imbalance should not throw you off balance: Choosing the right classifiers and performance metrics for brain decoding with imbalanced data. *NeuroImage* **2023**, *277*, 120253.
(18) Su, W.; Xie, X.-Q.; Liu, X.-W.; Gao, D.; Ma, C.-Y.; Zulfiqar, H.; Yang, H.; Lin, H.; Yu, X.-L.; Li, Y.-W. iRNA-ac4C: a novel computational method for effectively detecting N4-acetylcytidine sites in human mRNA. *Int. J. Biol. Macromol.* **2023**, *227*, 1174−1181.
(19) Dhakal, P.; Tayara, H.; Chong, K. T. An ensemble of stacking classifiers for improved prediction of miRNA−mRNA interactions. *Comput. Biol. Med.* **2023**, *164*, 107242.
(20) Xu, Y.; Zhao, X.; Liu, S.; Zhang, W. Predicting Long non-coding RNAs through feature ensemble learning. *Genomics* **2020**, *21*, 865.
(21) Noble, W. S.; Kuehn, S.; Thurman, R.; Yu, M.; Stamatoyannopoulos, J. Predicting the in vivo signature of human gene regulatory sequences. *Bioinformatics* **2005**, *21*, i338−i343.
(22) Lee, D.; Karchin, R.; Beer, M. A. Discriminative prediction of mammalian enhancers from DNA sequence. *Genome Res.* **2011**, *21*, 2167−2180.
(23) Lin, H.; Deng, E.-Z.; Ding, H.; Chen, W.; Chou, K.-C. iPro54-PseKNC: a sequence-based predictor for identifying sigma-54 promoters in prokaryote with pseudo k-tuple nucleotide composition. *Nucleic Acids Res.* **2014**, *42*, 12961−12972.
(24) Huang, T.; Li, S.; Jia, X.; Lu, H.; Liu, J. Neighbor2neighbor: Self-supervised denoising from single noisy images. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2021*; 2021, pp 14781−14790.
(25) Wang, P.; Fan, E.; Wang, P. Comparative analysis of image classification algorithms based on traditional machine learning and deep learning. *Pattern Recognit. Lett.* **2021**, *141*, 61−67.
(26) Cao, H.; Tan, C.; Gao, Z.; Xu, Y.; Chen, G.; Heng, P.-A.; Li, S. Z. A survey on generative diffusion models. *IEEE Trans. Knowl. Data Eng.* **2024**, *36*, 2814−2830.
(27) Wang, C.; Liu, Z.-P. Diffusion-based generation of gene regulatory networks from scRNA-seq data with DigNet. *Genome Res.* **2025**, *35*, 340−354.
(28) Wang, T.; Zhuo, L.; Chen, Y.; Fu, X.; Zeng, X.; Zou, Q. ECD-CDGI: An efficient energy-constrained diffusion model for cancer driver gene identification. *PLoS Comput. Biol.* **2024**, *20*, No. e1012400.
(29) Carlini, N.; Hayes, J.; Nasr, M.; Jagielski, M.; Sehwag, V.; Tramer, F.; Balle, B.; Ippolito, D.; Wallace, E. Extracting training data from diffusion models. In *32nd USENIX Security Symposium (USENIX Security 23), 2023*, 2023, pp 5253−5270.
(30) Zhao, Z.; Shi, G.; Wu, X.; Ren, R.; Gao, X.; Li, F. DIG-Mol: A Contrastive Dual-Interaction Graph Neural Network for Molecular Property Prediction. *IEEE J. Biomed. Health Inform.* **2025**, *29*, 1735−1746.
(31) He, W.; Jia, C.; Zou, Q. 4mCPred: machine learning methods for DNA N4-methylcytosine sites prediction. *Bioinformatics* **2019**, *35*, 593−601.
(32) Zuo, Y.; Zhang, B.; Dong, Y.; He, W.; Bi, Y.; Liu, X.; Zeng, X.; Deng, Z. Glypred: Lysine Glycation Site Prediction via CCU−LightGBM−BiLSTM Framework with Multi-Head Attention Mechanism. *J. Chem. Inf. Model.* **2024**, *64*, 6699−6711.
(33) Wu, T.; Huang, Q.; Liu, Z.; Wang, Y.; Lin, D. Distribution-balanced loss for multi-label classification in long-tailed datasets. In *Computer Vision−ECCV 2020:16th European Conference, Glasgow, UK, August 23−28, 2020, Proceedings, Part IV 16, 2020*; Springer, 2020, pp 162−178.
(34) Li, J.; Xiong, S.; Shi, H.; Cui, F.; Zhang, Z.; Wei, L. NeuroPred-AIMP: Multimodal Deep Learning for Neuropeptide Prediction via Protein Language Modeling and Temporal Convolutional Networks. *J. Chem. Inf. Model.* **2025**, *65*, 4740−4750.
(35) Tian, Y.; Zou, Q.; Wang, C.; Jia, C. Mamlcda: a meta-learning model for predicting circrna-disease association based on maml combined with cnn. *IEEE J. Biomed. Health Inform.* **2024**, *28*, 4325−4335.
(36) Li, X.; Xiong, H.; Li, X.; Wu, X.; Zhang, X.; Liu, J.; Bian, J.; Dou, D. Interpretable deep learning: Interpretation, interpretability, trustworthiness, and beyond. *Knowl. Inf. Syst.* **2022**, *64*, 3197−3234.
(37) Kang, B.; Garcia Garcia, D.; Lijffijt, J.; Santos-Rodríguez, R.; De Bie, T. Conditional t-SNE: more informative t-SNE embeddings. *Mach. Learn.* **2021**, *110*, 2905−2940.
(38) Xie, X.; Wu, C.; Ma, C.; Gao, D.; Su, W.; Huang, J.; Deng, K.; Yan, D.; Lin, H. Detecting key genes relative expression orderings as biomarkers for machine learning-based intelligent screening and analysis of type 2 diabetes mellitus. *Expert Syst. Appl.* **2024**, *255*, 124702.
(39) Lee, K. W.; Min, H. J.; Park, H. W.; Manavalan, B.; Jeon, Y.-J. Optimization of preprocessing strategies for developing AI-based disease diagnosis model using whole transcriptomic data. *Cancer Res.* **2025**, *85*, 2427.
(40) Zhang, Y.; Pan, X.; Shi, T.; Gu, Z.; Yang, Z.; Liu, M.; Xu, Y.; Yang, Y.; Ren, L.; Song, X.; et al. P450Rdb: a manually curated database of reactions catalyzed by cytochrome P450 enzymes. *J. Adv. Res.* **2024**, *63*, 35−42.

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